

BENTLEY
AutoPIPE Advanced 12.06.00.48

BENTLEY SYSTEMS, INCORPORATED
1065 N. PACIFIC CENTER DRIVE, SUITE 450
ANAHEIM, CA 92806

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**
** AUTOPIPE SYSTEM INFORMATION **
**

SYSTEM NAME : BBOU._ASME B31.3

PROJECT ID : BBOU

PREPARED BY : _____
I.O

CHECKED BY : _____
E N

1ST APPROVER : _____

2ND APPROVER : _____

PIPING CODE : ASME B31.3

YEAR : 2016

VERTICAL AXIS : Y

AMBIENT TEMPERATURE : 24.0 deg C

COMPONENT LIBRARY : AUTOPIPE

MATERIAL LIBRARY : B313-16

MODEL REVISION NUMBER : 26

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C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm) X	Y	Z	Mass Points
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*** SEGMENT C, LINE # BBOU-A-100_N9				
B05	43351.20	7500.00	-5300.00	0
C01	43351.20	8200.00	-5300.00	0

*** SEGMENT D, LINE # BBOU-8-P-014-C03				
C01	43351.20	8200.00	-5300.00	0
D01	43351.20	8220.00	-5300.00	0
D01 M	43351.20	8588.55	-5300.00	0
D02	43351.20	8957.11	-5300.00	0
D03 N	43351.20	9453.40	-5300.00	0
D03	43351.20	9758.20	-5300.00	0
D03 F	43351.20	9758.20	-5604.80	0
D10	43351.20	9758.20	-9300.00	0
D05 N	43351.20	9758.20	-12795.20	0
D05	43351.20	9758.20	-13100.00	0
D05 F	43351.20	9453.40	-13100.00	0
D06 N	43351.20	7804.80	-13100.00	0
D06	43351.20	7500.00	-13100.00	0
D06 F	43046.40	7500.00	-13100.00	0
D07	42471.20	7500.00	-13100.00	0
D07 M	42102.64	7500.00	-13100.00	0
D08	41734.09	7500.00	-13100.00	0
D09 N	40989.20	7500.00	-13100.00	0
D09	40684.40	7500.00	-13100.00	0
D09 F	40684.40	7500.00	-13404.80	0
E08	40684.40	7500.00	-14300.00	0

*** SEGMENT E, LINE # BBOU-24-P-013-C03				
AZ08	32348.40	7500.00	-5300.00	0
E01 N	32348.40	9985.60	-5300.00	0
E01	32348.40	10900.00	-5300.00	0
E01 F	32348.40	10900.00	-6214.40	0
E02	32348.40	10900.00	-9300.00	0
E03 N	32348.40	10900.00	-13385.60	0
E03	32348.40	10900.00	-14300.00	0
E03 F	32348.40	9985.60	-14300.00	0
E04 N	32348.40	8414.40	-14300.00	0
E04	32348.40	7500.00	-14300.00	0
E04 F	33262.80	7500.00	-14300.00	0
E05	35898.40	7500.00	-14300.00	0
E05 M	37041.40	7500.00	-14300.00	0
E06	38184.40	7500.00	-14300.00	0
E07	39684.40	7500.00	-14300.00	0
E08	40684.40	7500.00	-14300.00	0
E11	44484.40	7500.00	-14300.00	0
E09	47284.40	7500.00	-14300.00	0
E09 M	48427.40	7500.00	-14300.00	0
E10	49570.40	7500.00	-14300.00	0
T10	51570.20	7500.00	-14300.00	0

*** SEGMENT I, LINE # BBOU-A-100_N7				
B07	47751.20	7500.00	-5300.00	0

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C O O R D I N A T E S D A T A L I S T I N G

POINT	-----COORDINATE (mm)-----			Mass
NAME	X	Y	Z	Points

I01	47751.20	8200.00	-5300.00	0
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*** SEGMENT J, LINE # BBOU-4"-P-016-C06

I01	47751.20	8200.00	-5300.00	0
J01	47751.20	8220.00	-5300.00	0
J01 M	47751.20	8448.47	-5300.00	0
J02	47751.20	8676.95	-5300.00	0
J03 N	47751.20	8824.55	-5300.00	0
J03	47751.20	8976.95	-5300.00	0
J03 F	47751.20	8976.95	-5147.60	0
J04	47751.20	8976.95	-4500.00	0
J05 N	47751.20	8976.95	-3852.40	0
J05	47751.20	8976.95	-3700.00	0
J05 F	47598.80	8976.95	-3700.00	0
J06	42351.20	8976.95	-3700.00	0
J07	38351.20	8976.95	-3700.00	0
J08	33851.20	8976.95	-3700.00	0
J23	29174.70	8976.95	-3700.00	0
J09 N	24650.60	8976.95	-3700.00	0
J09	24498.20	8976.95	-3700.00	0
J09 F	24498.20	8976.95	-3852.40	0
J10	24498.20	8976.95	-7050.00	0
J11 N	24498.20	8976.95	-10347.60	0
J11	24498.20	8976.95	-10500.00	0
J11 F	24345.80	8976.95	-10500.00	0
J12	24098.20	8976.95	-10500.00	0
J13 N	22650.60	8976.95	-10500.00	0
J13	22498.20	8976.95	-10500.00	0
J13 F	22498.20	9129.35	-10500.00	0
J14 N	22498.20	9524.55	-10500.00	0
J14	22498.20	9676.95	-10500.00	0
J14 F	22345.80	9676.95	-10500.00	0
J15	22098.20	9676.95	-10500.00	0
J15 M	21869.73	9676.95	-10500.00	0
J16	21641.26	9676.95	-10500.00	0
J17	20741.25	9676.95	-10500.00	0
J18	20541.25	9676.95	-10500.00	0
J19 N	20493.65	9676.95	-10500.00	0
J19	20341.25	9676.95	-10500.00	0
J19 F	20341.26	9829.35	-10500.00	0
J20	20341.26	9876.95	-10500.00	0
J20 M	20341.26	9991.19	-10500.00	0
J21	20341.26	10105.43	-10500.00	0
J22	20341.26	10133.91	-10500.00	0

*** SEGMENT Q, LINE # BBOU-24-P-008-C03

T12	51570.20	7500.00	-11250.00	0
Q01	49570.20	7500.00	-11250.00	0
Q01 M	48427.20	7500.00	-11250.00	0
Q02	47284.20	7500.00	-11250.00	0
Q10	46084.20	7500.00	-11250.00	0
Q03	43884.20	7500.00	-11250.00	0

C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm)----- X	Y	Z	Mass Points
Q04	41901.20	7500.00	-11250.00	0
Q04 M	40758.20	7500.00	-11250.00	0
Q05	39615.20	7500.00	-11250.00	0
Q11	37215.20	7500.00	-11250.00	0
Q06 N	37129.60	7500.00	-11250.00	0
Q06	36215.20	7500.00	-11250.00	0
Q06 F	36215.20	8414.40	-11250.00	0
Q07 N	36215.20	9985.60	-11250.00	0
Q07	36215.20	10900.00	-11250.00	0
Q07 F	36215.20	10900.00	-10335.60	0
Q08 N	36215.20	10900.00	-9164.40	0
Q08	36215.20	10900.00	-8250.00	0
Q08 F	36215.20	9985.60	-8250.00	0
P03	36215.20	7500.00	-8250.00	0

*** SEGMENT T, LINE # BBOU-24-P-008-C03

T00	34558.20	7500.00	-19350.00	0
T01	36098.20	7500.00	-19350.00	0
T02	37098.20	7500.00	-19350.00	0
T02 M	38241.20	7500.00	-19350.00	0
T03	39384.20	7500.00	-19350.00	0
T16	41734.20	7500.00	-19350.00	0
T04	44084.20	7500.00	-19350.00	0
T04 M	45227.20	7500.00	-19350.00	0
T05	46370.20	7500.00	-19350.00	0
T06	47370.20	7500.00	-19350.00	0
T07	49620.20	7500.00	-19350.00	0
T08 N	50655.80	7500.00	-19350.00	0
T08	51570.20	7500.00	-19350.00	0
T08 F	51570.20	7500.00	-18435.60	0
T14	51570.20	7500.00	-17350.00	0
T10	51570.20	7500.00	-14300.00	0
T15	51570.20	7500.00	-12775.00	0
T12	51570.20	7500.00	-11250.00	0
T13	51570.20	7500.00	-10250.00	0

*** SEGMENT U, LINE # BBOU-24-P-008-C03 BYPASS

T06	47370.20	7500.00	-19350.00	0
U01 N	47370.20	7500.00	-18264.40	0
U01	47370.20	7500.00	-17350.00	0
U01 F	46455.80	7500.00	-17350.00	0
U02	45670.20	7500.00	-17350.00	0
U02 M	44527.20	7500.00	-17350.00	0
U03	43384.20	7500.00	-17350.00	0
U06	39741.20	7500.00	-17350.00	0
U04 N	37012.60	7500.00	-17350.00	0
U04	36098.21	7500.00	-17350.00	0
U04 F	36098.20	7500.00	-18264.39	0
T01	36098.20	7500.00	-19350.00	0

*** SEGMENT X, LINE # BBOU-24-P-003-C03

T00	34558.20	7500.00	-19350.00	0
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C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm)-----			Mass Points
	X	Y	Z	

X01	33018.20	7500.00	-19350.00	0
X01 M	31875.20	7500.00	-19350.00	0
X02	30732.20	7500.00	-19350.00	0
X03	29032.20	7500.00	-19350.00	0
Y04	27332.20	7500.00	-19350.00	0

*** SEGMENT Y, LINE # BBOU-24-P-003-C03

Y00	16412.20	7500.00	-24350.00	0
Y01	16412.20	7500.00	-21750.00	0
Y05	16412.20	7500.00	-20550.00	0
Y02 N	16412.20	7500.00	-20264.40	0
Y02	16412.20	7500.00	-19350.00	0
Y02 F	17326.60	7500.00	-19350.00	0
Y03	19612.20	7500.00	-19350.00	0
Y06	23472.20	7500.00	-19350.00	0
Y04	27332.20	7500.00	-19350.00	0

*** SEGMENT Z, LINE # BBOU-24-PF-XXXX-120-E

Y00	16412.20	7500.00	-24350.00	0
Z01	16412.20	7500.00	-25650.00	0
Z02	16412.20	7500.00	-26950.00	0

*** SEGMENT AA, LINE # BBOU-6-B-005-B06

J22	20341.26	10133.91	-10500.00	0
AA01	20274.31	10133.91	-10500.00	0
AA02	20154.31	10133.91	-10500.00	0
AA03	19294.31	10133.91	-10500.00	0
AA04	18834.31	10133.91	-10500.00	0
AA04M	18529.26	10133.91	-10500.00	0
AA05	18224.20	10133.91	-10500.00	0
AA06	17824.20	10133.91	-10500.00	0
AA07	17424.20	10133.91	-10500.00	0
AA08N	17252.80	10133.91	-10500.00	0
AA08	17024.20	10133.91	-10500.00	0
AA08F	17024.20	9905.31	-10500.00	0
AW01	17024.20	9733.91	-10500.00	0

*** SEGMENT AB, LINE # BBOU-4"-P-016-C06_B

J18	20541.25	9676.95	-10500.00	0
AB01N	20541.25	9676.95	-10352.40	0
AB01	20541.25	9676.95	-10200.00	0
AB01F	20388.85	9676.95	-10200.00	0
AB02	19591.26	9676.95	-10200.00	0
AB02M	19362.78	9676.95	-10200.00	0
AB03	19134.31	9676.95	-10200.00	0
AB04	18677.11	9676.95	-10200.00	0

*** SEGMENT AD, LINE # BBOU-6-B-005-B06_B

AB04	18677.11	9676.95	-10200.00	0
AD01	18050.65	9676.95	-10200.00	0
AD02N	17576.60	9676.95	-10200.00	0
AD02	17424.20	9676.95	-10200.00	0

C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm) X	Y	Z	Mass Points
AD02F	17424.20	9829.35	-10200.00	0
AD03N	17424.20	9981.51	-10200.00	0
AD03	17424.20	10133.91	-10200.00	0
AD03F	17424.20	10133.91	-10352.40	0
AA07	17424.20	10133.91	-10500.00	0

*** SEGMENT AE, LINE # BBOU-A-300-N7

O03	44191.20	7500.00	-8250.00	0
AE01	44191.20	8200.00	-8250.00	0

*** SEGMENT AF, LINE # BBOU-4"-P-021-C06

AE01	44191.20	8200.00	-8250.00	0
AF01	44191.20	8220.00	-8250.00	0
AF01M	44191.20	8448.47	-8250.00	0
AF02	44191.20	8676.95	-8250.00	0
AF03N	44191.20	8824.55	-8250.00	0
AF03	44191.20	8976.95	-8250.00	0
AF03F	44191.20	8976.95	-8097.60	0
AF04	44191.20	8976.95	-7530.00	0
AF05	44191.20	8976.95	-4850.00	0
AF06N	44191.20	8976.95	-4202.40	0
AF06	44191.20	8976.95	-4050.00	0
AF06F	44038.80	8976.95	-4050.00	0
AF07	42291.20	8976.95	-4050.00	0
AF08	38291.20	8976.95	-4050.00	0
AF09	33791.20	8976.95	-4050.00	0
AF27	29319.70	8976.95	-4050.00	0
AF10N	25000.60	8976.95	-4050.00	0
AF10	24848.20	8976.95	-4050.00	0
AF10F	24848.20	8976.95	-4202.40	0
AF11	24848.20	8976.95	-7050.00	0
AF12N	24848.20	8976.95	-10897.60	0
AF12	24848.20	8976.95	-11050.00	0
AF12F	24695.80	8976.95	-11050.00	0
AF13	24098.20	8976.95	-11050.00	0
AF14N	22650.60	8976.95	-11050.00	0
AF14	22498.20	8976.95	-11050.00	0
AF14F	22498.20	8976.95	-11202.40	0
AF15	22498.20	8976.95	-11650.00	0
AF16N	22498.20	8976.95	-11897.60	0
AF16	22498.20	8976.95	-12050.00	0
AF16F	22498.20	9129.35	-12050.00	0
AF17N	22498.20	9524.55	-12050.00	0
AF17	22498.20	9676.95	-12050.00	0
AF17F	22345.80	9676.95	-12050.00	0
AF18	22198.20	9676.95	-12050.00	0
AF19	22098.20	9676.95	-12050.00	0
AF19M	21869.73	9676.95	-12050.00	0
AF20	21641.26	9676.95	-12050.00	0
AF21	20741.25	9676.95	-12050.00	0
AF22	20541.25	9676.95	-12050.00	0
AF23N	20493.65	9676.95	-12050.00	0

C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm) X	Y	Z	Mass Points
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AF23	20341.25	9676.95	-12050.00	0
AF23F	20341.26	9829.35	-12050.00	0
AF24	20341.26	9876.95	-12050.00	0
AF24M	20341.26	9991.19	-12050.00	0
AF25	20341.26	10105.43	-12050.00	0
AF26	20341.26	10133.91	-12050.00	0

*** SEGMENT AI, LINE # BBOU-XX-B-007-B06

AF26	20341.26	10133.91	-12050.00	0
AI01	20274.31	10133.91	-12050.00	0
AI02	20154.31	10133.91	-12050.00	0
AI03	19294.31	10133.91	-12050.00	0
AI04	18834.31	10133.91	-12050.00	0
AI04M	18529.26	10133.91	-12050.00	0
AI05	18224.20	10133.91	-12050.00	0
AI06	17824.20	10133.91	-12050.00	0
AI07	17424.20	10133.91	-12050.00	0
AI08N	17252.80	10133.91	-12050.00	0
AI08	17024.20	10133.91	-12050.00	0
AI08F	17024.20	9905.31	-12050.00	0
AW02	17024.20	9733.91	-12050.00	0

*** SEGMENT AJ, LINE # BBOU-4"-P-021-C06_B

AF22	20541.25	9676.95	-12050.00	0
AJ01N	20541.25	9676.95	-11902.40	0
AJ01	20541.25	9676.95	-11750.00	0
AJ01F	20388.85	9676.95	-11750.00	0
AJ02	19591.26	9676.95	-11750.00	0
AJ02M	19362.78	9676.95	-11750.00	0
AJ03	19134.31	9676.95	-11750.00	0
AJ04	18677.11	9676.95	-11750.00	0

*** SEGMENT AK, LINE # BBOU-XX-B-007-B06_B

AJ04	18677.11	9676.95	-11750.00	0
AK01	18050.65	9676.95	-11750.00	0
AK02N	17576.60	9676.95	-11750.00	0
AK02	17424.20	9676.95	-11750.00	0
AK02F	17424.20	9829.35	-11750.00	0
AK03N	17424.20	9981.51	-11750.00	0
AK03	17424.20	10133.91	-11750.00	0
AK03F	17424.20	10133.91	-11902.40	0
AI07	17424.20	10133.91	-12050.00	0

*** SEGMENT AR, LINE # BBOU-6-P-006-C06

T07	49620.20	7500.00	-19350.00	0
AR01N	49620.20	8748.40	-19350.00	0
AR01	49620.20	8977.00	-19350.00	0
AR01F	49620.20	8977.00	-19121.40	0
AR02	49620.20	8977.00	-18350.00	0
AR03N	49620.20	8977.00	-15841.60	0
AR03	49620.20	8977.00	-15613.00	0
AR03F	49391.61	8977.00	-15613.00	0

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C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm)-----			Mass Points
	X	Y	Z	
AR04	48020.21	8977.00	-15613.00	0
AR05	44020.21	8977.00	-15613.00	0
AR06	38020.21	8977.00	-15613.00	0
AR22	33506.96	8977.00	-15613.00	0
AR21	28993.71	8977.00	-15613.00	0
AR07	24967.21	8977.00	-15613.00	0
AR08N	24326.81	8977.00	-15613.00	0
AR08	24098.21	8977.00	-15613.00	0
AR08F	24098.21	8977.00	-15384.40	0
AR09N	24098.21	8977.00	-13878.60	0
AR09	24098.21	8977.00	-13650.00	0
AR09F	23869.61	8977.00	-13650.00	0
AR10N	23326.80	8977.00	-13650.00	0
AR10	23098.21	8977.00	-13650.00	0
AR10F	23098.21	9205.60	-13650.00	0
AR11N	23098.21	9448.40	-13650.00	0
AR11	23098.21	9677.00	-13650.00	0
AR11F	22869.61	9677.00	-13650.00	0
AR12	22698.21	9677.00	-13650.00	0
AR13	22398.21	9677.00	-13650.00	0
AR13M	22093.15	9677.00	-13650.00	0
AR14	21788.10	9677.00	-13650.00	0
AR15	21248.10	9677.00	-13650.00	0
AR16	20708.10	9677.00	-13650.00	0
AR16M	20403.04	9677.00	-13650.00	0
AR17	20097.99	9677.00	-13650.00	0
AR18N	19876.59	9677.00	-13650.00	0
AR18	19647.99	9677.00	-13650.00	0
AR18F	19647.99	9905.60	-13650.00	0
AR19	19647.99	9977.00	-13650.00	0
AR19M	19647.99	10091.50	-13650.00	0
AR20	19647.99	10206.00	-13650.00	0

*** SEGMENT AS, LINE # BBOU-6-B-003-B06

AR20	19647.99	10206.00	-13650.00	0
AS01	19297.99	10206.00	-13650.00	0
AS02	18947.99	10206.00	-13650.00	0
AS02M	18642.93	10206.00	-13650.00	0
AS03	18337.88	10206.00	-13650.00	0
AS04N	17252.80	10206.00	-13650.00	0
AS04	17024.20	10206.00	-13650.00	0
AS04F	17024.20	9977.40	-13650.00	0
AW04	17024.20	9733.91	-13650.00	0

*** SEGMENT AT, LINE # BBOU-6-P-007-C06

AR12	22698.21	9677.00	-13650.00	0
AT01N	22698.21	9677.00	-14091.40	0
AT01	22698.21	9677.00	-14320.00	0
AT01F	22469.61	9677.00	-14320.00	0
AT02	22448.21	9677.00	-14320.00	0
AT03	22148.21	9677.00	-14320.00	0
AT03M	21843.15	9677.00	-14320.00	0

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C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm) X	Y	----- Z	Mass Points
AT04	21538.10	9677.00	-14320.00	0
AT05	21248.10	9677.00	-14320.00	0
AT06	20708.10	9677.00	-14320.00	0
AT06M	20403.04	9677.00	-14320.00	0
AT07	20097.99	9677.00	-14320.00	0
AT08N	19876.59	9677.00	-14320.00	0
AT08	19647.99	9677.00	-14320.00	0
AT08F	19647.99	9905.60	-14320.00	0
AT09	19647.99	9977.00	-14320.00	0
AT09M	19647.99	10091.50	-14320.00	0
AT10	19647.99	10206.00	-14320.00	0

*** SEGMENT AV, LINE # BBOU-6-B-004-B06

AT10	19647.99	10206.00	-14320.00	0
AV01	19297.99	10206.00	-14320.00	0
AV02	18947.99	10206.00	-14320.00	0
AV02M	18642.93	10206.00	-14320.00	0
AV03	18337.88	10206.00	-14320.00	0
AV04N	17252.80	10206.00	-14320.00	0
AV04	17024.20	10206.00	-14320.00	0
AV04F	17024.20	9977.40	-14320.00	0
AW05	17024.20	9733.91	-14320.00	0

*** SEGMENT AW, LINE # BBOU-6-B-001-B06

AW00	17024.20	9733.91	-10200.00	0
AW01	17024.20	9733.91	-10500.00	0
AW02	17024.20	9733.91	-12050.00	0
AW03	17024.20	9733.91	-12588.00	0
AW04	17024.20	9733.91	-13650.00	0
AW05	17024.20	9733.91	-14320.00	0
AW06N	17024.20	9733.91	-15191.40	0
AW06	17024.20	9733.91	-15420.00	0
AW06F	17024.20	9505.31	-15420.00	0
AW07N	17024.20	8802.51	-15420.00	0
AW07	17024.20	8573.91	-15420.00	0
AW07F	16795.60	8573.91	-15420.00	0
AW08	16224.20	8573.91	-15420.00	0
AW09	11924.20	8573.91	-15420.00	0
AW10N	10952.80	8573.91	-15420.00	0
AW10	10724.20	8573.91	-15420.00	0
AW10F	10724.20	8573.91	-15191.40	0
AW11	10724.20	8573.91	-13420.00	0
AW12N	10724.20	8573.91	-11648.60	0
AW12	10724.20	8573.91	-11420.00	0
AW12F	10724.20	8345.31	-11420.00	0
AW13	10724.20	4573.91	-11420.00	0

*** SEGMENT AX, LINE # BBOU-6-B-007-B06

AX00	10724.20	4573.91	-8820.00	0
AX01	10724.20	4593.91	-8820.00	0
AX01M	10724.20	4822.51	-8820.00	0
AX02	10724.20	5051.11	-8820.00	0

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C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm)----- X Y Z	Mass Points
AX03N	10724.20 8345.31 -8820.00	0
AX03	10724.20 8573.91 -8820.00	0
AX03F	10724.20 8573.91 -8591.40	0
AX04	10724.20 8573.91 -7620.00	0
AX05	10724.20 8573.91 -2220.00	0
AX06	10724.20 8573.91 3780.00	0
AX07	10724.20 8573.91 5480.00	0

*** SEGMENT AY, LINE # BBOU-8-P-008-C03

O02	48391.20	7500.00	-8250.00	0
AY01	48391.20	8200.00	-8250.00	0
AY01M	48391.20	8568.55	-8250.00	0
AY02	48391.20	8937.11	-8250.00	0
AY03N	48391.20	9433.40	-8250.00	0
AY03	48391.20	9738.20	-8250.00	0
AY03F	48391.20	9738.20	-8554.80	0
AY04N	48391.20	9738.20	-9645.20	0
AY04	48391.20	9738.20	-9950.00	0
AY04F	48391.20	9433.40	-9950.00	0
AY05N	48391.20	7804.80	-9950.00	0
AY05	48391.20	7500.00	-9950.00	0
AY05F	48086.40	7500.00	-9950.00	0
AY06	47541.20	7500.00	-9950.00	0
AY07	46541.20	7500.00	-9950.00	0
AY07M	46172.65	7500.00	-9950.00	0
AY08	45804.09	7500.00	-9950.00	0
AY09N	44189.00	7500.00	-9950.00	0
AY09	43884.20	7500.00	-9950.00	0
AY09F	43884.20	7500.00	-10254.80	0
Q03	43884.20	7500.00	-11250.00	0

Pipe ID/ Material CladMaterial ---Line Class---	Nom/ Sch	O.D. mm	---Thickness(mm)-----				Spec		InsuDen/ LingDen/ kg/m3	Weight(N/m) Pipe/ Cont	Ling/ Insu/ Clad	Total	E/ W/ f	Composition/ Steel Group/ Tcr
Tag No. : <None> 8"C03 API5L-B 900	200 120	219.07	18.26	3.00	2.28	0	0	0.70	0.000 0.000 0.000	885 180	0 0 0	1065	1.00 1.00 1.00	Carbon Steel Carbon 0.000
Tag No. : <None> 24"C06 API5L-X60 900	600 NS	609.60	38.89	6.00	4.86	0	0	0.70	0.000 0.000 0.000	5356 1524	0 0 0	6880	1.00 1.00 1.00	Carbon Steel Carbon 0.000
Tag No. : <None> 4"C06 API5L-B 900	100 XXS	114.30	17.12	6.00	2.14	0	0	0.70	0.000 0.000 0.000	401 34.54	0 0 0	436	1.00 1.00 1.00	Carbon Steel Carbon 0.000
Tag No. : <None> 6"B06 API5L-B 150	150 XS	168.27	10.97	6.00	1.37	0	0	0.70	0.000 0.000 0.000	417 115	0 0 0	532	1.00 1.00 1.00	Carbon Steel Carbon 0.000
Tag No. : <None> 4"B06 API5L-B 150	100 120	114.30	11.13	6.00	1.39	0	0	0.70	0.000 0.000 0.000	277 45.66	0 0 0	323	1.00 1.00 1.00	Carbon Steel Carbon 0.000
Tag No. : <None> 6"C06 API5L-B 900	150 160	168.27	18.26	6.00	2.28	0	0	0.70	0.000 0.000 0.000	661 93.54	0 0 0	755	1.00 1.00 1.00	Carbon Steel Carbon 0.000
Tag No. : <None> 24C70 A790-S31803 900	600 NS	609.60	24.20	0	3.03	0	0	0.70	0.000 0.000 0.000	3419 1697	0 0 0	5116	1.00 1.00 1.00	Austenitic Stainless 0.000

M A T E R I A L D A T A L I S T I N G

Material Name	Pipe ID	Density kg/m3	Pois. Ratio	Temper. deg C	Modulus E6 N/mm2 Axial	Modulus E6 N/mm2 Hoop	Modulus E6 N/mm2 Shear	Expans. mm/m	Composition
API5L-B	8"C03	7833.0	0.30	24.0 100.0 -29.0 23.0	0.20320 0.19816 0.20593 0.20327	0.20320	0.07815	0.8900 -0.5971 -0.0115	Carbon Steel
API5L-X60	24"C06	7833.0	0.30	24.0 100.0 -29.0 23.0 12.7	0.20320 0.19816 0.20593 0.20327 0.20382	0.20320	0.07815	0.8900 -0.5971 -0.0115 -0.1280	Carbon Steel
API5L-B	4"C06	7833.0	0.30	24.0 100.0 -29.0 23.0	0.20320 0.19816 0.20593 0.20327	0.20320	0.07815	0.8900 -0.5971 -0.0115	Carbon Steel
A790-S31803	24C70	7833.0	0.30	24.0 100.0 -29.0 23.0	0.19493 0.18980 0.19847 0.19500	0.19493	0.07497	0.9733 -0.6863 -0.0150	Austenitic Stainless
API5L-B	4"B06	7833.0	0.30	24.0 60.0 -29.0 23.0	0.20320 0.20080 0.20593 0.20327	0.20320	0.07815	0.4154 -0.5971 -0.0115	Carbon Steel
API5L-B	6"B06	7833.0	0.30	24.0 60.0 -29.0 23.0 12.7	0.20320 0.20080 0.20593 0.20327 0.20382	0.20320	0.07815	0.4154 -0.5971 -0.0115 -0.1280	Carbon Steel
API5L-B	6"C06	7833.0	0.30	24.0 100.0 -29.0 23.0	0.20320 0.19816 0.20593 0.20327	0.20320	0.07815	0.8900 -0.5971 -0.0115	Carbon Steel

M A T E R I A L A L L O W A B L E D A T A L I S T I N G

Material Name	Pipe ID	Temper. deg C	Allow. N/mm2	Yield N/mm2	Ultimate Weld	Reduction Factor W
API5L-B	8"C03	24.0	137.9	241.3	413.7	1.000
		100.0	137.9	220.4		1.000
		-29.0	137.9	241.3		1.000
		23.0	137.9	241.3		1.000
API5L-X60	24"C06	24.0	172.4	413.7	517.1	1.000
		100.0	172.4	0.0		1.000
		-29.0	172.4	0.0		1.000
		23.0	172.4	0.0		1.000
API5L-B	4"C06	24.0	137.9	241.3	413.7	1.000
		100.0	137.9	220.4		1.000
		-29.0	137.9	241.3		1.000
		23.0	137.9	241.3		1.000
A790-S31803	24C70	24.0	206.8	448.2	620.5	1.000
		100.0	205.9	395.1		1.000
		-29.0	206.8	448.2		1.000
		23.0	206.8	448.2		1.000
API5L-B	4"B06	24.0	137.9	241.3	413.7	1.000
		60.0	137.9	233.3		1.000
		-29.0	137.9	241.3		1.000
		23.0	137.9	241.3		1.000
API5L-B	6"B06	24.0	137.9	241.3	413.7	1.000
		60.0	137.9	233.3		1.000
		-29.0	137.9	241.3		1.000
		23.0	137.9	241.3		1.000
API5L-B	6"C06	24.0	137.9	241.3	413.7	1.000
		100.0	137.9	220.4		1.000
		-29.0	137.9	241.3		1.000
		23.0	137.9	241.3		1.000

POINT NAME	CASE	PRESS. N/mm2	TEMPER deg C	EXPAN. mm/m	MODULUS E6 N/mm2	ALLOW STRESS	YIELD STRESS
*** SEGMENT C, LINE # BBOU-A-100 N9							
B05	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
C01	Same as previous point.						
*** SEGMENT D, LINE # BBOU-8-P-014-C03							
C01	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
E08	Same as previous point.						
*** SEGMENT E, LINE # BBOU-24-P-013-C03							
AZ08	T1	11.0000	100	0.890	0.19816	172.37	0
	T2	11.0000	-29.00	-0.597	0.20593	172.37	0
	T3	11.0000	23.00	-0.012	0.20327	172.37	0
T10	Same as previous point.						
*** SEGMENT I, LINE # BBOU-A-100 N7							
B07	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
I01	Same as previous point.						
*** SEGMENT J, LINE # BBOU-4"-P-016-C06							
I01	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
J22	Same as previous point.						
*** SEGMENT Q, LINE # BBOU-24-P-008-C03							
T12	T1	11.0000	100	0.890	0.19816	172.37	0
	T2	11.0000	-29.00	-0.597	0.20593	172.37	0
	T3	11.0000	23.00	-0.012	0.20327	172.37	0
P03	Same as previous point.						
*** SEGMENT T, LINE # BBOU-24-P-008-C03							
T00	T1	11.0000	100	0.890	0.19816	172.37	0
	T2	11.0000	-29.00	-0.597	0.20593	172.37	0
	T3	11.0000	23.00	-0.012	0.20327	172.37	0
T13	Same as previous point.						
*** SEGMENT U, LINE # BBOU-24-P-008-C03 BYPASS							
T06	T1	11.0000	100	0.890	0.19816	172.37	0
	T2	11.0000	-29.00	-0.597	0.20593	172.37	0
	T3	11.0000	23.00	-0.012	0.20327	172.37	0
T01	Same as previous point.						
*** SEGMENT X, LINE # BBOU-24-P-003-C03							
T00	T1	11.0000	100	0.890	0.19816	172.37	0
	T2	11.0000	-29.00	-0.597	0.20593	172.37	0
	T3	11.0000	23.00	-0.012	0.20327	172.37	0
Y04	Same as previous point.						
*** SEGMENT Y, LINE # BBOU-24-P-003-C03							
Y00	T1	11.0000	100	0.973	0.18980	205.93	395.12
	T2	11.0000	-29.00	-0.686	0.19847	206.84	448.16
	T3	11.0000	23.00	-0.015	0.19500	206.84	448.16
Y04	Same as previous point.						

POINT NAME	CASE	PRESS. N/mm2	TEMPER deg C	EXPAN. mm/m	MODULUS E6 N/mm2	ALLOW STRESS	YIELD STRESS
***	SEGMENT Z, LINE # BBOU-24-PF-XXXX-120-E						
Y00	T1	11.0000	100	0.973	0.18980	205.93	395.12
	T2	11.0000	-29.00	-0.686	0.19847	206.84	448.16
	T3	11.0000	23.00	-0.015	0.19500	206.84	448.16
Z02	Same as previous point.						
***	SEGMENT AA, LINE # BBOU-6-B-005-B06						
J22	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
AW01	Same as previous point.						
***	SEGMENT AB, LINE # BBOU-4"-P-016-C06_B						
J18	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
AB04	Same as previous point.						
***	SEGMENT AD, LINE # BBOU-6-B-005-B06_B						
AB04	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
AA07	Same as previous point.						
***	SEGMENT AE, LINE # BBOU-A-300-N7						
O03	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
AE01	Same as previous point.						
***	SEGMENT AF, LINE # BBOU-4"-P-021-C06						
AE01	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
AF26	Same as previous point.						
***	SEGMENT AI, LINE # BBOU-XX-B-007-B06						
AF26	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
AW02	Same as previous point.						
***	SEGMENT AJ, LINE # BBOU-4"-P-021-C06_B						
AF22	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
AJ04	Same as previous point.						
***	SEGMENT AK, LINE # BBOU-XX-B-007-B06_B						
AJ04	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
AI07	Same as previous point.						
***	SEGMENT AR, LINE # BBOU-6-P-006-C06						
T07	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
AR20	Same as previous point.						
***	SEGMENT AS, LINE # BBOU-6-B-003-B06						

OPERATING TEMPERATURE AND PRESSURE DATA
STRESSES IN N/mm2

POINT NAME	CASE	PRESS. N/mm2	TEMPER deg C	EXPAN. mm/m	MODULUS E6 N/mm2	ALLOW STRESS	YIELD STRESS
AR20	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
AW04	Same as previous point.						
*** SEGMENT AT, LINE # BBOU-6-P-007-C06							
AR12	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
AT10	Same as previous point.						
*** SEGMENT AV, LINE # BBOU-6-B-004-B06							
AT10	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
AW05	Same as previous point.						
*** SEGMENT AW, LINE # BBOU-6-B-001-B06							
AW00	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
AW13	Same as previous point.						
*** SEGMENT AX, LINE # BBOU-6-B-007-B06							
AX00	T1	1.9600	60.00	0.415	0.20080	137.90	233.32
	T2	1.9600	-29.00	-0.597	0.20593	137.90	241.32
	T3	1.9600	23.00	-0.012	0.20327	137.90	241.32
AX07	Same as previous point.						
*** SEGMENT AY, LINE # BBOU-8-P-008-C03							
O02	T1	11.0000	100	0.890	0.19816	137.90	220.41
	T2	11.0000	-29.00	-0.597	0.20593	137.90	241.32
	T3	11.0000	23.00	-0.012	0.20327	137.90	241.32
Q03	Same as previous point.						

u User-defined value

* Non-code material for allowable stress;
Non-standard material for expansion and modulus

S E G M E N T D A T A L I S T I N G

Segment Name	First Node	Last Node	Line Number	Apply Wind	Apply Bowling	Apply Buoyancy
C	B05	C01	BBOU-A-100 N9	Yes	No	No
D	C01	E08	BBOU-8-P-014-C03	Yes	No	No
E	AZ08	T10	BBOU-24-P-013-C03	Yes	No	No
I	B07	I01	BBOU-A-100 N7	Yes	No	No
J	I01	J22	BBOU-4"-P-016-C06	Yes	No	No
Q	T12	P03	BBOU-24-P-008-C03	Yes	No	No
T	T00	T13	BBOU-24-P-008-C03	Yes	No	No
U	T06	T01	BBOU-24-P-008-C03_BYPASS	Yes	No	No
X	T00	Y04	BBOU-24-P-003-C03	Yes	No	No
Y	Y00	Y04	BBOU-24-P-003-C03	Yes	No	No
Z	Y00	Z02	BBOU-24-PF-XXXX-120-E	Yes	No	No
AA	J22	AW01	BBOU-6-B-005-B06	Yes	No	No
AB	J18	AB04	BBOU-4"-P-016-C06_B	Yes	No	No
AD	AB04	AA07	BBOU-6-B-005-B06_B	Yes	No	No
AE	O03	AE01	BBOU-A-300-N7	Yes	No	No
AF	AE01	AF26	BBOU-4"-P-021-C06	Yes	No	No
AI	AF26	AW02	BBOU-XX-B-007-B06	Yes	No	No
AJ	AF22	AJ04	BBOU-4"-P-021-C06_B	Yes	No	No
AK	AJ04	AI07	BBOU-XX-B-007-B06_B	Yes	No	No
AR	T07	AR20	BBOU-6-P-006-C06	Yes	No	No
AS	AR20	AW04	BBOU-6-B-003-B06	Yes	No	No
AT	AR12	AT10	BBOU-6-P-007-C06	Yes	No	No
AV	AT10	AW05	BBOU-6-B-004-B06	Yes	No	No
AW	AW00	AW13	BBOU-6-B-001-B06	Yes	No	No
AX	AX00	AX07	BBOU-6-B-007-B06	Yes	No	No
AY	O02	Q03	BBOU-8-P-008-C03	Yes	No	No

L O A D S S U M M A R Y D A T A L I S T I N G

WIND LOAD CASES :

Number of load cases : 4

Ground elevation for wind : 2000.00 mm

Wind shape factor multiplier for piping: 1.000

Wind shape factor multiplier for beams: 1.000

Exposure factor for buried pipes : 0.000

Exposure factor for submerged pipes : 0.000

Exposed Segments - C D E I J Q T U X Y Z AA AB AD AE

 AF AI AJ AK AR AS AT AV AW AX AY

Wind application method : Projected

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 1 - W1

Direction - Global X-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 1 - W1

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 2 - W2

Direction - Global -X-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 2 - W2

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 3 - W3

Direction - Global Z-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 3 - W3

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 4 - W4

Direction - Global -Z-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 4 - W4

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

S U P P O R T D A T A L I S T I N G

Point Name	Support Type	Support ID	Conn.to /Dir	Comp.Wt (kg)	Stiff.	Gap 1 (mm)	Gap 2 (mm)	Fric. Fact.	GapSet /#hgr	Preload (N)	Ld.Var	Size	Figure
Tag No.: <None>													
D10	Guide-V	D10	1	Ground	0.000	0.000	0.0	3.0	0.10	Weightless			
	Guide-H	D10	1	Ground	0.000	0.000	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
E02	Guide-V	E02	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	E02	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
E05	M Shoe	E05	M1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
E11	Guide-V	E11	1	Ground	0.000	0.000	0.0	3.0	0.10	Weightless			
	Guide-H	E11	1	Ground	0.000	0.000	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
E09	M Shoe	E09	M1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
J04	Guide-V	J04	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	J04	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
J06	Guide-V	J06	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	J06	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
J07	Guide-V	J07	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	J07	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
J07	Linestop	J07	2	Ground	0.000	Rigid	0.0	0.0					
Attachment ID: <None>													
Tag No.: <None>													
J08	Guide-V	J08	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	J08	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
J23	Guide-V	J23	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	J23	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
J10	Guide-V	J10	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	J10	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
J12	Guide-V	J12	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	J12	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Attachment ID: <None>													
Tag No.: <None>													
Q01	M Shoe	Q01	M1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			

S U P P O R T D A T A L I S T I N G

Point Name	Support Type	Support ID	Conn.to /Dir	Comp.Wt (kg)	Stiff.	Gap 1 (mm)	Gap 2 (mm)	Fric. Fact.	GapSet /#hgr	Preload (N)	Ld.Var	Size	Figure
Tag No.: <None>													
Attachment ID: <None>													
Q10	Guide-V	Q10	1	Ground	0.000	0.000	0.0	3.0	0.10	Weightless			
	Guide-H	Q10	1	Ground	0.000	0.000	3.0	3.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
Q04	M Shoe	Q04	M1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
Q11	Guide-V	Q11	1	Ground	0.000	0.000	0.0	3.0	0.10	Weightless			
	Guide-H	Q11	1	Ground	0.000	0.000	3.0	3.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
T02	M Shoe	T02	M1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
T16	Guide-V	T16	1	Ground	0.000	0.000	0.0	3.0	0.10	Weightless			
	Guide-H	T16	1	Ground	0.000	0.000	3.0	3.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
T04	M Shoe	T04	M1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
T14	Shoe	T14	1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
U02	M Shoe	U02	M1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
X01	M Shoe	X01	M1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
Y05	Shoe	Y05	1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
Y03	Shoe	Y03	1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
Y06	Guide-V	Y06	1	Ground	0.000	0.000	0.0	3.0	0.10	Weightless			
	Guide-H	Y06	1	Ground	0.000	0.000	3.0	3.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
AA03	Guide-V	AA03	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	AA03	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
AA06	Guide-V	AA06	1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless			
	Guide-H	AA06	1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless			
Tag No.: <None>													
Attachment ID: <None>													
AD01	Shoe	AD01	1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless			

S U P P O R T D A T A L I S T I N G

Point Name	Support Type	Support ID	Conn.to /Dir	Comp.Wt (kg)	Stiff.	Gap 1 (mm)	Gap 2 (mm)	Fric. Fact.	GapSet /#hgr	Preload (N)	Ld.Var	Size	Figure
Tag No.: <None>						Attachment ID: <None>							
AF04	Guide-V	AF04 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AF04 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AF07	Guide-V	AF07 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AF07 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AF08	Guide-V	AF08 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AF08 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AF08	Linestop	AF08 2	Ground	0.000	Rigid	0.0	0.0						
Tag No.: <None>						Attachment ID: <None>							
AF09	Guide-V	AF09 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AF09 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AF27	Guide-V	AF27 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AF27 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AF11	Guide-V	AF11 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AF11 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AF13	Guide-V	AF13 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AF13 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AI03	Guide-V	AI03 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AI03 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AI06	Guide-V	AI06 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AI06 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AK01	Shoe	AK01 1	Ground	0.000	Rigid	0.0	2540.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AR02	Guide-V	AR02 1	Ground	0.000	0.000	0.0	3.0	0.10		Weightless			
	Guide-H	AR02 1	Ground	0.000	0.000	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AR04	Guide-V	AR04 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AR04 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None>						Attachment ID: <None>							
AR05	Guide-V	AR05 1	Ground	0.000	0.000	0.0	3.0	0.10		Weightless			

S U P P O R T D A T A L I S T I N G

Point Name	Support Type	Support ID	Conn.to /Dir	Comp.Wt (kg)	Stiff.	Gap 1 (mm)	Gap 2 (mm)	Fric. Fact.	GapSet /#hgr	Preload (N)	Ld.Var	Size	Figure
	Guide-H	AR05 1	Ground	0.000	0.000	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AR06	Guide-V	AR06 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AR06 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AR06	Linestop	AR06 2	Ground	0.000	0.000	0.0	0.0						
Tag No.: <None> Attachment ID: <None>													
AR22	Guide-V	AR22 1	Ground	0.000	0.000	0.0	3.0	0.10		Weightless			
	Guide-H	AR22 1	Ground	0.000	0.000	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AR21	Guide-V	AR21 1	Ground	0.000	0.000	0.0	3.0	0.10		Weightless			
	Guide-H	AR21 1	Ground	0.000	0.000	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AR07	Guide-V	AR07 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AR07 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AS01	Guide-V	AS01 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AS01 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AV01	Guide-V	AV01 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AV01 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AW08	Guide-V	AW08 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AW08 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AW09	Shoe	AW09 1	Ground	0.000	Rigid	0.0	2540.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AW11	Guide-V	AW11 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AW11 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AX04	Guide-V	AX04 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AX04 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AX05	Shoe	AX05 2	Ground	0.000	Rigid	0.0	2540.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AX06	Guide-V	AX06 1	Ground	0.000	Rigid	0.0	3.0	0.10		Weightless			
	Guide-H	AX06 1	Ground	0.000	Rigid	3.0	3.0	0.10		Weightless			
Tag No.: <None> Attachment ID: <None>													
AY06	Shoe	AY06 1	Ground	0.000	Rigid	0.0	2540.0	0.10		Weightless			

Spring Manufacturer: Anvil/Grinnell

NOTE 1: No soil supports present in the system.

Gap 1 : V-stop,Guide-V=down, Linestop,Incline,Tie/link=backward, Guide-H=Left

Gap 2 : V-stop,Guide-V=Up , Linestop,Incline,Tie/link=forward , Guide-H=Right

Stiffness units for rotation support: N.m/deg , all others: N/mm

B E N D D A T A L I S T I N G

Point Name	Bend Type	Radius (mm)	Angle (deg)	OD (mm)	tnom (mm)	Material	Flan^ /Cuts	SIF in	SIF out	Analysis Set	Flexibility (K)	Pressure Case
D03	Elbow	305L	90.0	219.1	18.3	API5L-B	0/0	1.34	1.11	1	2.99	None
								1.34	1.11	2	2.99	None
D05	Elbow	305L	90.0	219.1	18.3	API5L-B	0/0	1.34	1.11	1	2.99	None
								1.34	1.11	2	2.99	None
D06	Elbow	305L	90.0	219.1	18.3	API5L-B	0/0	1.34	1.11	1	2.99	None
								1.34	1.11	2	2.99	None
D09	Elbow	305L	90.0	219.1	18.3	API5L-B	0/0	1.34	1.11	1	2.99	None
								1.34	1.11	2	2.99	None
E01	Elbow	914L	90.0	609.6	38.9	API5L-X60	0/0	1.56	1.30	1	3.78	None
								1.56	1.30	2	3.78	None
E03	Elbow	914L	90.0	609.6	38.9	API5L-X60	0/0	1.56	1.30	1	3.78	None
								1.56	1.30	2	3.78	None
E04	Elbow	914L	90.0	609.6	38.9	API5L-X60	0/0	1.56	1.30	1	3.78	None
								1.56	1.30	2	3.78	None
J03	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J05	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J09	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J11	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J13	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J14	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
J19	Elbow	152L	90.0	114.3	17.1	API5L-B	1/0	1.00	1.00	1	1.52	None
								1.00	1.00	2	1.52	None
Q06	Elbow	914L	90.0	609.6	38.9	API5L-X60	0/0	1.56	1.30	1	3.78	None
								1.56	1.30	2	3.78	None
Q07	Elbow	914L	90.0	609.6	38.9	API5L-X60	0/0	1.56	1.30	1	3.78	None
								1.56	1.30	2	3.78	None
Q08	Elbow	914L	90.0	609.6	38.9	API5L-X60	0/0	1.56	1.30	1	3.78	None
								1.56	1.30	2	3.78	None
T08	Elbow	914L	90.0	609.6	38.9	API5L-X60	0/0	1.56	1.30	1	3.78	None
								1.56	1.30	2	3.78	None
U01	Elbow	914L	90.0	609.6	38.9	API5L-X60	0/0	1.56	1.30	1	3.78	None
								1.56	1.30	2	3.78	None
U04	Elbow	914L	90.0	609.6	38.9	API5L-X60	0/0	1.56	1.30	1	3.78	None
								1.56	1.30	2	3.78	None
Y02	Elbow	914L	90.0	609.6	24.2	A790-S31803	0/0	2.22	1.85	1	6.39	None
								2.22	1.85	2	6.39	None
AA08	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
AB01	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
AD02	Elbow	152L	90.0	114.3	11.1	API5L-B	0/0	1.22	1.01	1	2.59	None
								1.22	1.01	2	2.59	None
AD03	Elbow	152L	90.0	114.3	11.1	API5L-B	0/0	1.22	1.01	1	2.59	None
								1.22	1.01	2	2.59	None
AF03	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None

B E N D D A T A L I S T I N G

Point Name	Bend Type	Radius (mm)	Angle (deg)	OD (mm)	tnom (mm)	Material	Flan^ /Cuts	SIF in	SIF out	Analysis Set	Flexibility (K)	Pressure Case
AF06	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
AF10	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
AF12	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
AF14	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
AF16	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
AF17	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
AF23	Elbow	152L	90.0	114.3	17.1	API5L-B	1/0	1.00	1.00	1	1.52	None
								1.00	1.00	2	1.52	None
AI08	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
AJ01	Elbow	152L	90.0	114.3	17.1	API5L-B	0/0	1.00	1.00	1	1.49	None
								1.00	1.00	2	1.49	None
AK02	Elbow	152L	90.0	114.3	11.1	API5L-B	0/0	1.22	1.01	1	2.59	None
								1.22	1.01	2	2.59	None
AK03	Elbow	152L	90.0	114.3	11.1	API5L-B	0/0	1.22	1.01	1	2.59	None
								1.22	1.01	2	2.59	None
AR01	Elbow	229L	90.0	168.3	18.3	API5L-B	0/0	1.10	1.00	1	2.22	None
								1.10	1.00	2	2.22	None
AR03	Elbow	229L	90.0	168.3	18.3	API5L-B	0/0	1.10	1.00	1	2.22	None
								1.10	1.00	2	2.22	None
AR08	Elbow	229L	90.0	168.3	18.3	API5L-B	0/0	1.10	1.00	1	2.22	None
								1.10	1.00	2	2.22	None
AR09	Elbow	229L	90.0	168.3	18.3	API5L-B	0/0	1.10	1.00	1	2.22	None
								1.10	1.00	2	2.22	None
AR10	Elbow	229L	90.0	168.3	18.3	API5L-B	0/0	1.10	1.00	1	2.22	None
								1.10	1.00	2	2.22	None
AR11	Elbow	229L	90.0	168.3	18.3	API5L-B	0/0	1.10	1.00	1	2.22	None
								1.10	1.00	2	2.22	None
AR18	Elbow	229L	90.0	168.3	18.3	API5L-B	1/0	1.04	1.00	1	2.12	None
								1.04	1.00	2	2.12	None
AS04	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
AT01	Elbow	229L	90.0	168.3	18.3	API5L-B	0/0	1.10	1.00	1	2.22	None
								1.10	1.00	2	2.22	None
AT08	Elbow	229L	90.0	168.3	18.3	API5L-B	1/0	1.04	1.00	1	2.12	None
								1.04	1.00	2	2.12	None
AV04	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
AW06	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
AW07	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
AW10	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
AW12	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None

B E N D D A T A L I S T I N G

Point Name	Bend Type	Radius (mm)	Angle (deg)	OD (mm)	tnom (mm)	Material	Flan^ /Cuts	SIF in	SIF out	Analysis Set	Flexibility (K)	Pressure Case
AX03	Elbow	229L	90.0	168.3	11.0	API5L-B	0/0	1.64	1.37	1	4.07	None
								1.64	1.37	2	4.07	None
AY03	Elbow	305L	90.0	219.1	18.3	API5L-B	0/0	1.34	1.11	1	2.99	None
								1.34	1.11	2	2.99	None
AY04	Elbow	305L	90.0	219.1	18.3	API5L-B	0/0	1.34	1.11	1	2.99	None
								1.34	1.11	2	2.99	None
AY05	Elbow	305L	90.0	219.1	18.3	API5L-B	0/0	1.34	1.11	1	2.99	None
								1.34	1.11	2	2.99	None
AY09	Elbow	305L	90.0	219.1	18.3	API5L-B	0/0	1.34	1.11	1	2.99	None
								1.34	1.11	2	2.99	None

^ = Number of bend ends where either a flange or valve is within distance
(L/D) x Nominal diameter, where L/D is defined under Tools > Model options > Edit
L = Long radius

T E E D A T A L I S T I N G

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	**Crotch Radius (mm)	*Crotch Thick (mm)	SIF in	SIF out
E08	E-Header D-Branch	609.6 219.1	38.9 18.3	API5L-X60 API5L-B	Welding			1.45	1.60
T10	T-Header E-Branch	609.6 609.6	38.9 38.9	API5L-X60 API5L-X60	Welding			1.45	1.60
J18	J-Header AB-Branch	114.3 114.3	17.1 17.1	API5L-B API5L-B	Other			1.00	1.00
T12	T-Header Q-Branch	609.6 609.6	38.9 38.9	API5L-X60 API5L-X60	Welding			1.45	1.60
Q03	Q-Header AY-Branch	609.6 219.1	38.9 18.3	API5L-X60 API5L-B	Welding			1.45	1.60
T01	T-Header U-Branch	609.6 609.6	38.9 38.9	API5L-X60 API5L-X60	Welding			1.45	1.60
T06	T-Header U-Branch	609.6 609.6	38.9 38.9	API5L-X60 API5L-X60	Welding			1.45	1.60
T07	T-Header AR-Branch	609.6 168.3	38.9 18.3	API5L-X60 API5L-B	Fitting			1.53	1.53
AA07	AA-Header AD-Branch	168.3 114.3	11.0 11.1	API5L-B API5L-B	Welding			1.43	1.57
AW01	AW-Header AA-Branch	168.3 168.3	11.0 11.0	API5L-B API5L-B	Welding			1.43	1.57
AF22	AF-Header AJ-Branch	114.3 114.3	17.1 17.1	API5L-B API5L-B	Welding			1.00	1.00
AI07	AI-Header AK-Branch	168.3 114.3	11.0 11.1	API5L-B API5L-B	Welding			1.43	1.57
AW02	AW-Header AI-Branch	168.3 168.3	11.0 11.0	API5L-B API5L-B	Welding			1.43	1.57
AR12	AR-Header AT-Branch	168.3 168.3	18.3 18.3	API5L-B API5L-B	Welding			1.06	1.09
AW04	AW-Header AS-Branch	168.3 168.3	11.0 11.0	API5L-B API5L-B	Welding			1.43	1.57
AW05	AW-Header AV-Branch	168.3 168.3	11.0 11.0	API5L-B API5L-B	Welding			1.43	1.57

* = Pad thickness for reinforced tee.

**= Fillet radius for Raised tee or discontinuity dist. for Thickened tee.

V A L V E D A T A L I S T I N G

Point Name	Type	Dnom (mm)	Length (mm)	Valve Weight (kg)	Auto	Symbol	Actuator Weight (kg)	DX (mm)	DY (mm)	DZ (mm)	SAF	Joint Type	SIF	Size /Avg. (mm)	Type /Max. (mm)
Tag No. : MV107 D01	CATALOG	200	737.11	579	Yes	No					0.00	WN	1.00		
Tag No. : MV1062 D07	CATALOG	200	737.11	579	Yes	No					0.00	WN	1.00		
Tag No. : MV1017 E05	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1017 E09	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1072 J01	CATALOG	100	456.95	145	Yes	No					2.00	WN	1.00		
Tag No. : MV1044 J15	CATALOG	100	456.95	145	Yes	No					2.00	WN	1.00		
Tag No. : PSV1003 J20	CATALOG	100	228.48	145	Yes	No					2.00	WN	1.00		
Tag No. : MV1017 Q01	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1059 Q04	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1043 T02	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1045 T04	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1044 U02	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : SDV1001 X01	CATALOG	600	2286.00	6786	Yes	No					0.00	WN	1.00		
Tag No. : MV1066 AA04	CATALOG	150	610.11	359	Yes	No					0.00	WN	1.00		
Tag No. : MV1065 AB02	CATALOG	100	456.95	145	Yes	No					2.00	WN	1.00		
Tag No. : E4RNC AB03	CATALOG	100	457.20	193	Yes	No					5.40	WN	1.00		
Tag No. : MV1072 AF01	CATALOG	100	456.95	145	Yes	No					2.00	WN	1.00		

V A L V E D A T A L I S T I N G

Point Name	Type	Dnom (mm)	Length (mm)	Valve Weight (kg)	Auto	Symbol	Actuator Weight (kg)	DX (mm)	DY (mm)	DZ (mm)	SAF	Joint Type	SIF	Size /Avg. (mm)	Type /Max. (mm)
Tag No. : MV1044 AF19	CATALOG	100	456.95	145	Yes	No						2.00 WN	1.00		
Tag No. : PSV1003 AF24	CATALOG	100	228.48	145	Yes	No						2.00 WN	1.00		
Tag No. : MV1066 AI04	CATALOG	150	610.11	359	Yes	No						0.00 WN	1.00		
Tag No. : MV1065 AJ02	CATALOG	100	456.95	145	Yes	No						2.00 WN	1.00		
Tag No. : E4RNC AJ03	CATALOG	100	457.20	193	Yes	No						5.40 WN	1.00		
Tag No. : MV101 N0 AR13	CATALOG	150	610.11	359	Yes	No						0.00 WN	1.00		
Tag No. : MV101 N0 AR16	CATALOG	150	610.11	359	Yes	No						0.00 WN	1.00		
Tag No. : PSV1003 AR19	CATALOG	150	229.00	359	Yes	No						0.00 WN	1.00		
Tag No. : MV1011 AS02	CATALOG	150	610.11	359	Yes	No						0.00 WN	1.00		
Tag No. : MV101 N0 AT03	CATALOG	150	610.11	359	Yes	No						0.00 WN	1.00		
Tag No. : MV101 N0 AT06	CATALOG	150	610.11	359	Yes	No						0.00 WN	1.00		
Tag No. : PSV1003 AT09	CATALOG	150	229.00	359	Yes	No						0.00 WN	1.00		
Tag No. : MV1011 AV02	CATALOG	150	610.11	359	Yes	No						0.00 WN	1.00		
Tag No. : MV1011 AX01	CATALOG	150	457.20	153	Yes	No						2.00 WN	1.00		
Tag No. : MV1011 AY01	CATALOG	200	737.11	579	Yes	No						0.00 WN	1.00		
Tag No. : MV1011 AY07	CATALOG	200	737.11	579	Yes	No						0.00 WN	1.00		

C A T A L O G V A L V E D A T A L I S T I N G

Point Name : D01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : D07
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : E05
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : E09
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : J01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : J15
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : J20
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

C A T A L O G V A L V E D A T A L I S T I N G

Point Name : Q01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : Q04
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : T02
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : T04
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : U02
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : X01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AA04
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

C A T A L O G V A L V E D A T A L I S T I N G

Point Name : AB02
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AB03
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Gate Valves
Type : Gate Valves
Main End : FL
Rating : 900

Point Name : AF01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AF19
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AF24
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AI04
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AJ02
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

C A T A L O G V A L V E D A T A L I S T I N G

Point Name : AJ03
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Gate Valves
Type : Gate Valves
Main End : FL
Rating : 900

Point Name : AR13
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AR16
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AR19
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AS02
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AT03
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AT06
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

C A T A L O G V A L V E D A T A L I S T I N G

Point Name : AT09
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AV02
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AX01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 150

Point Name : AY01
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : AY07
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

F L A N G E D A T A L I S T I N G											
Point	FLG		Dnom		Flange	Bolt			Size	Type	ANSI
Name	No.	Type	(mm)	Rating	Weight	Weight	Joint	SIF	/Avg.	/Max.	Check
					(kg)	(kg)	Type		(mm)	(mm)	
Tag No. :		<None>									
D01	1	WELDNECK	200	900	168.74	84.37	WN	1.00			No
Tag No. :		<None>									
D02	1	WELDNECK	200	900	168.74	84.37	WN	1.00			No
Tag No. :		<None>									
D07	1	WELDNECK	200	150	39.01	0.00	WN	1.00			No
Tag No. :		<None>									
D08	1	WELDNECK	200	150	39.01	0.00	WN	1.00			No
Tag No. :		<None>									
E05	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
E06	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
E09	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
E10	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
J01	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
J02	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
J15	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
J16	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
J20	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
J22	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
Q01	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
Q02	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
Q04	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
Q05	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
T02	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
T03	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
T04	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
T05	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
U02	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
U03	1	WELDNECK	600	900	1764.00	0.00	WN	1.00			No
Tag No. :		<None>									
T00	1	WELDNECK	600	900	1764.00	882.00	WN	1.00			No
Tag No. :		<None>									
T00	2	WELDNECK	600	900	1764.00	882.00	WN	1.00			No

F L A N G E D A T A L I S T I N G

Point Name	FLG No.	Type	Dnom (mm)	Rating	Flange Weight (kg)	Bolt Weight (kg)	Joint Type	SIF	Size /Avg. (mm)	Type /Max. (mm)	ANSI Check
Tag No. :		<None>									
X01	1	WELDNECK	600		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
X02	1	WELDNECK	600		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
Y01	1	WELDNECK	600		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
Y01	2	WELDNECK	600		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
Y04	1	WELDNECK	600		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
Y04	2	WELDNECK	600		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
Y00	1	WELDNECK	600		9001764.00	0.00	WN	1.00			No
Tag No. :		<None>									
Y00	2	WELDNECK	600		9001764.00	0.00	WN	1.00			No
Tag No. :		<None>									
Z02	1	WELDNECK	600		9001764.00	0.00	WN	1.00			No
Tag No. :		<None>									
AA04	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No
Tag No. :		<None>									
AA05	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No
Tag No. :		<None>									
AB02	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AB03	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AB03	2	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AB04	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AF01	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AF02	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AF19	1	WELDNECK	100	900	23.13	11.56	WN	1.00			No
Tag No. :		<None>									
AF20	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AF20	2	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AF24	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AF26	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AI04	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No
Tag No. :		<None>									
AI05	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No
Tag No. :		<None>									
AJ02	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No
Tag No. :		<None>									
AJ03	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No

F L A N G E					D A T A		L I S T I N G					
Point Name	FLG No.	Type	Dnom (mm)	Rating	Flange Weight (kg)	Bolt Weight (kg)	Joint Type	SIF	Size /Avg. (mm)	Type /Max. (mm)	ANSI Check	
Tag No. :		<None>										
AJ03	2	WELDNECK	100	900	23.13	0.00	WN	1.00			No	
Tag No. :		<None>										
AJ04	1	WELDNECK	100	900	23.13	0.00	WN	1.00			No	
Tag No. :		<None>										
AR13	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AR14	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AR16	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AR17	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AR19	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AR20	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AS02	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No	
Tag No. :		<None>										
AS03	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No	
Tag No. :		<None>										
AT03	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AT04	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AT06	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AT07	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AT09	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AT10	1	WELDNECK	150	900	49.90	0.00	WN	1.00			No	
Tag No. :		<None>										
AV02	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No	
Tag No. :		<None>										
AV03	1	WELDNECK	150	300	18.60	0.00	WN	1.00			No	
Tag No. :		<None>										
AX01	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No	
Tag No. :		<None>										
AX02	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No	
Tag No. :		<None>										
AX07	1	WELDNECK	150	150	11.34	0.00	WN	1.00			No	
Tag No. :		<None>										
AY01	1	WELDNECK	200	150	39.01	0.00	WN	1.00			No	
Tag No. :		<None>										
AY02	1	WELDNECK	200	150	39.01	0.00	WN	1.00			No	
Tag No. :		<None>										
AY07	1	WELDNECK	200	150	39.01	0.00	WN	1.00			No	
Tag No. :		<None>										
AY08	1	WELDNECK	200	150	39.01	0.00	WN	1.00			No	

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R E D U C E R D A T A L I S T I N G

From Point	To Point	Length (mm)	From OD (mm)	To OD (mm)	From Thk (mm)	To Thk (mm)	From Material	Cone Angle (deg)	SIF
AA01	AA02	120.00	114.3	168.3	11.1	11.0	API5L-B		1.00
AI01	AI02	120.00	114.3	168.3	11.1	11.0	API5L-B		1.00

C E N T E R O F G R A V I T Y R E P O R T

Description	Weight (kg)	X CoG (mm)	Y CoG (mm)	Z CoG (mm)
-----	-----	-----	-----	-----
Valve	: 61505.44	40293.48	7693.81	-15297.89
Support	: 0.00	0.00	0.00	0.00
Flange	: 53124.40	33831.73	7560.87	-17954.50
Flexible Joint	: 0.00	0.00	0.00	0.00
Additional Weight	: 0.00	0.00	0.00	0.00
Pipes (Run, Bend, Tee, Reducer)	: 63917.41	36970.93	8222.15	-14217.11
PipeTot (Valve+Support+Flange+Flex. Jt.+Add. Wt.+Pipes):	178547.17	37181.46	7843.40	-15701.46
Insulation	: 0.00	0.00	0.00	0.00
Cladding	: 0.00	0.00	0.00	0.00
Lining	: 0.00	0.00	0.00	0.00
Contents	: 21295.92	36624.13	8041.31	-15093.61
Beam	: 0.00	0.00	0.00	0.00
PipeTot + Ins.	: 178547.17	37181.46	7843.40	-15701.46
PipeTot + Ins. + Clad.	: 178547.17	37181.46	7843.40	-15701.46
PipeTot + Ins. + Clad. + Lin.	: 178547.17	37181.46	7843.40	-15701.46
PipeTot + Ins. + Clad. + Lin. + Cont.	: 199843.02	37122.11	7864.48	-15636.66
PipeTot + Beam	: 178547.17	37181.46	7843.40	-15701.46
PipeTot + Ins. + Beam	: 178547.17	37181.46	7843.40	-15701.46
PipeTot + Ins. + Clad. + Beam	: 178547.17	37181.46	7843.40	-15701.46
PipeTot + Ins. + Clad. + Lin. + Beam	: 178547.17	37181.46	7843.40	-15701.46
PipeTot + Ins. + Clad. + Lin. + Cont. + Beam	: 199843.02	37122.11	7864.48	-15636.66

NOTE: Fluid Density Factor is assumed as 1.0.

NOTE: COG report only includes the currently visible beams and segments.

All pipe segments are included.

No beams present in the model.

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L O A D C A S E		D E S C R I P T I O N
Load Case	Load Case Name	Description
-----	-----	-----
GR	Gravity	
T1	Thermal 1	
T2	Thermal 2	
T3	Thermal 3	
W1	Wind 1	
W2	Wind 2	
W3	Wind 3	
W4	Wind 4	
P1	Press 1	
P2	Press 2	
P3	Press 3	

HYDROTEST LOAD SUMMARY

Specific Gravity of Fluid : 1.000
Hydrotest Pressure Case : P3
Hydrotest Pressure Factor : 1.500
Hydrotest Temperature Case : T3
Hydrotest Temperature Factor : 1.000
Cut-short Case : None
Include Insulation & Cladding: No

Point Summary

Pipe	Number of Points in the System			Mass	Pipe	Weight of System(kg)	
	Frame	Soil	Total			Contents	Total
452	0	0	452	0	178547.2	21295.9	199843.1

ANCHOR DATA LISTING

Point Name	Translational Stiffness			Rotational Stiffness			Release Hanger		
	KTX N/mm	KTY N/mm	KTZ N/mm	KRX N.m/deg	KRY N.m/deg	KRZ N.m/deg	X XX	Y YY	Z ZZ
B05	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
AZ08	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
B07	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
J17	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
P03	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
T15	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
X03	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
Z01	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
O03	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
AF21	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
AR15	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
AT05	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			
AW13	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			

ANCHOR DATA LISTING

Point Name	Translational Stiffness			Rotational Stiffness			Release Hanger		
	KTX N/mm	KTY N/mm	KTZ N/mm	KRX N.m/deg	KRY N.m/deg	KRZ N.m/deg	X XX	Y YY	Z ZZ
AX01	RIGID	RIGID	RIGID	Connected to: Ground RIGID RIGID RIGID					
AX07	RIGID	RIGID	RIGID	Connected to: Ground RIGID RIGID RIGID					
002	RIGID	RIGID	RIGID	Connected to: Ground RIGID RIGID RIGID					

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X	Y	Z
*** Segment C										
Origin	B05	Point	8"C03	0.00	43351.20	7500.00	-5300.00	43351.20	7500.00	-5300.00
B05	C01	Run	8"C03	58.33	0.00	700.00	0.00	43351.20	8200.00	-5300.00
*** Segment D										
Origin	C01	Point	8"C03	0.00	43351.20	8200.00	-5300.00	43351.20	8200.00	-5300.00
C01	D01	Junction	8"C03	1.67	0.00	20.00	0.00	43351.20	8220.00	-5300.00
D01	D02	Valve	8"C03	61.43	0.00	737.11	0.00	43351.20	8957.11	-5300.00
D02	D03	Bend	8"C03	66.76	0.00	801.09	0.00	43351.20	9758.20	-5300.00
D03	D10	Run	8"C03	333.33	0.00	0.00	-4000.00	43351.20	9758.20	-9300.00
D10	D05	Bend	8"C03	316.67	0.00	0.00	-3800.00	43351.20	9758.20	-13100.00
D05	D06	Bend	8"C03	188.18	0.00	-2258.20	0.00	43351.20	7500.00	-13100.00
D06	D07	Run	8"C03	73.33	-880.00	0.00	0.00	42471.20	7500.00	-13100.00
D07	D08	Valve	8"C03	61.43	-737.11	0.00	0.00	41734.09	7500.00	-13100.00
D08	D09	Bend	8"C03	87.47	-1049.69	0.00	0.00	40684.40	7500.00	-13100.00
D09	E08	Tee	8"C03	100.00	0.00	0.00	-1200.00	40684.40	7500.00	-14300.00
*** Segment E										
Origin	AZ08	Point	24"C06	0.00	32348.40	7500.00	-5300.00	32348.40	7500.00	-5300.00
AZ08	E01	Bend	24"C06	283.33	0.00	3400.00	0.00	32348.40	10900.00	-5300.00
E01	E02	Run	24"C06	333.33	0.00	0.00	-4000.00	32348.40	10900.00	-9300.00
E02	E03	Bend	24"C06	416.67	0.00	0.00	-5000.00	32348.40	10900.00	-14300.00
E03	E04	Bend	24"C06	283.33	0.00	-3400.00	0.00	32348.40	7500.00	-14300.00
E04	E05	Run	24"C06	295.83	3550.00	0.00	0.00	35898.40	7500.00	-14300.00
E05	E06	Valve	24"C06	190.50	2286.00	0.00	0.00	38184.40	7500.00	-14300.00
E06	E07	Run	24"C06	125.00	1500.00	0.00	0.00	39684.40	7500.00	-14300.00
E07	E08	Tee	24"C06	83.33	1000.00	0.00	0.00	40684.40	7500.00	-14300.00
E08	E11	Run	24"C06	316.67	3800.00	0.00	0.00	44484.40	7500.00	-14300.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm)		Z
								X	Y	
E11	E09	Run	24"C06	233.33	2800.00	0.00	0.00	47284.40	7500.00	-14300.00
E09	E10	Valve	24"C06	190.50	2286.00	0.00	0.00	49570.40	7500.00	-14300.00
E10	T10	Tee	24"C06	166.65	1999.80	0.00	0.00	51570.20	7500.00	-14300.00
*** Segment I										
Origin	B07	Point	4"C06	0.00	47751.20	7500.00	-5300.00	47751.20	7500.00	-5300.00
B07	I01	Run	4"C06	58.33	0.00	700.00	0.00	47751.20	8200.00	-5300.00
*** Segment J										
Origin	I01	Point	4"C06	0.00	47751.20	8200.00	-5300.00	47751.20	8200.00	-5300.00
I01	J01	Junction	4"C06	1.67	0.00	20.00	0.00	47751.20	8220.00	-5300.00
J01	J02	Valve	4"C06	38.08	0.00	456.95	0.00	47751.20	8676.95	-5300.00
J02	J03	Bend	4"C06	25.00	0.00	300.00	0.00	47751.20	8976.95	-5300.00
J03	J04	Run	4"C06	66.67	0.00	0.00	800.00	47751.20	8976.95	-4500.00
J04	J05	Bend	4"C06	66.67	0.00	0.00	800.00	47751.20	8976.95	-3700.00
J05	J06	Run	4"C06	450.00	-5400.00	0.00	0.00	42351.20	8976.95	-3700.00
J06	J07	Run	4"C06	333.33	-4000.00	0.00	0.00	38351.20	8976.95	-3700.00
J07	J08	Run	4"C06	375.00	-4500.00	0.00	0.00	33851.20	8976.95	-3700.00
J08	J23	Run	4"C06	389.71	-4676.50	0.00	0.00	29174.70	8976.95	-3700.00
J23	J09	Bend	4"C06	389.71	-4676.50	0.00	0.00	24498.20	8976.95	-3700.00
J09	J10	Run	4"C06	279.17	0.00	0.00	-3350.00	24498.20	8976.95	-7050.00
J10	J11	Bend	4"C06	287.50	0.00	0.00	-3450.00	24498.20	8976.95	-10500.00
J11	J12	Run	4"C06	33.33	-400.00	0.00	0.00	24098.20	8976.95	-10500.00
J12	J13	Bend	4"C06	133.33	-1600.00	0.00	0.00	22498.20	8976.95	-10500.00
J13	J14	Bend	4"C06	58.33	0.00	700.00	0.00	22498.20	9676.95	-10500.00
J14	J15	Run	4"C06	33.33	-400.00	0.00	0.00	22098.20	9676.95	-10500.00
J15	J16	Valve	4"C06	38.08	-456.95	0.00	0.00	21641.26	9676.95	-10500.00
J16	J17	Run	4"C06	75.00	-900.00	0.00	0.00	20741.25	9676.95	-10500.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X Y		Z
J17	J18	Tee	4"C06	16.67	-200.00	0.00	0.00	20541.25	9676.95	-10500.00
J18	J19	Bend	4"C06	16.67	-200.00	0.00	0.00	20341.25	9676.95	-10500.00
J19	J20	Run	4"C06	16.67	0.00	200.00	0.00	20341.26	9876.95	-10500.00
J20	J21	Valve	4"C06	19.04	0.00	228.48	0.00	20341.26	10105.43	-10500.00
J21	J22	Run	4"C06	2.37	0.00	28.48	0.00	20341.26	10133.91	-10500.00
*** Segment Q										
Origin	T12	Tee	24"C06	0.00	51570.20	7500.00	-11250.00	51570.20	7500.00	-11250.00
T12	Q01	Run	24"C06	166.67	-2000.00	0.00	0.00	49570.20	7500.00	-11250.00
Q01	Q02	Valve	24"C06	190.50	-2286.00	0.00	0.00	47284.20	7500.00	-11250.00
Q02	Q10	Run	24"C06	100.00	-1200.00	0.00	0.00	46084.20	7500.00	-11250.00
Q10	Q03	Tee	24"C06	183.33	-2200.00	0.00	0.00	43884.20	7500.00	-11250.00
Q03	Q04	Run	24"C06	165.25	-1983.00	0.00	0.00	41901.20	7500.00	-11250.00
Q04	Q05	Valve	24"C06	190.50	-2286.00	0.00	0.00	39615.20	7500.00	-11250.00
Q05	Q11	Run	24"C06	200.00	-2400.00	0.00	0.00	37215.20	7500.00	-11250.00
Q11	Q06	Bend	24"C06	83.33	-1000.00	0.00	0.00	36215.20	7500.00	-11250.00
Q06	Q07	Bend	24"C06	283.33	0.00	3400.00	0.00	36215.20	10900.00	-11250.00
Q07	Q08	Bend	24"C06	250.00	0.00	0.00	3000.00	36215.20	10900.00	-8250.00
Q08	P03	Run	24"C06	283.33	0.00	-3400.00	0.00	36215.20	7500.00	-8250.00
*** Segment T										
Origin	T00	Point	24"C06	0.00	34558.20	7500.00	-19350.00	34558.20	7500.00	-19350.00
T00	T01	Tee	24"C06	128.33	1540.00	0.00	0.00	36098.20	7500.00	-19350.00
T01	T02	Run	24"C06	83.33	1000.00	0.00	0.00	37098.20	7500.00	-19350.00
T02	T03	Valve	24"C06	190.50	2286.00	0.00	0.00	39384.20	7500.00	-19350.00
T03	T16	Run	24"C06	195.83	2350.00	0.00	0.00	41734.20	7500.00	-19350.00
T16	T04	Run	24"C06	195.83	2350.00	0.00	0.00	44084.20	7500.00	-19350.00
T04	T05	Valve	24"C06	190.50	2286.00	0.00	0.00	46370.20	7500.00	-19350.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X	Y	Z
T05	T06	Tee	24"C06	83.33	1000.00	0.00	0.00	47370.20	7500.00	-19350.00
T06	T07	Tee	24"C06	187.50	2250.00	0.00	0.00	49620.20	7500.00	-19350.00
T07	T08	Bend	24"C06	162.50	1950.00	0.00	0.00	51570.20	7500.00	-19350.00
T08	T14	Run	24"C06	166.67	0.00	0.00	2000.00	51570.20	7500.00	-17350.00
T14	T10	Tee	24"C06	254.17	0.00	0.00	3050.00	51570.20	7500.00	-14300.00
T10	T15	Run	24"C06	127.08	0.00	0.00	1525.00	51570.20	7500.00	-12775.00
T15	T12	Tee	24"C06	127.08	0.00	0.00	1525.00	51570.20	7500.00	-11250.00
T12	T13	Run	24"C06	83.33	0.00	0.00	1000.00	51570.20	7500.00	-10250.00
*** Segment U										
Origin	T06	Tee	24"C06	0.00	47370.20	7500.00	-19350.00	47370.20	7500.00	-19350.00
T06	U01	Bend	24"C06	166.67	0.00	0.00	2000.00	47370.20	7500.00	-17350.00
U01	U02	Run	24"C06	141.67	-1700.00	0.00	0.00	45670.20	7500.00	-17350.00
U02	U03	Valve	24"C06	190.50	-2286.00	0.00	0.00	43384.20	7500.00	-17350.00
U03	U06	Run	24"C06	303.58	-3643.00	0.00	0.00	39741.20	7500.00	-17350.00
U06	U04	Bend	24"C06	303.58	-3643.00	0.00	0.00	36098.21	7500.00	-17350.00
U04	T01	Tee	24"C06	166.67	-0.01	0.00	-2000.00	36098.20	7500.00	-19350.00
*** Segment X										
Origin	T00	Point	24"C06	0.00	34558.20	7500.00	-19350.00	34558.20	7500.00	-19350.00
T00	X01	Junction	24"C06	128.33	-1540.00	0.00	0.00	33018.20	7500.00	-19350.00
X01	X02	Valve	24"C06	190.50	-2286.00	0.00	0.00	30732.20	7500.00	-19350.00
X02	X03	Run	24"C06	141.67	-1700.00	0.00	0.00	29032.20	7500.00	-19350.00
X03	Y04	Run	24"C06	141.67	-1700.00	0.00	0.00	27332.20	7500.00	-19350.00
*** Segment Y										
Origin	Y00	Point	24C70	0.00	16412.20	7500.00	-24350.00	16412.20	7500.00	-24350.00
Y00	Y01	Junction	24C70	216.67	0.00	0.00	2600.00	16412.20	7500.00	-21750.00
Y01	Y05	Run	24C70	100.00	0.00	0.00	1200.00	16412.20	7500.00	-20550.00

From	To	Comp.	Pipe Id	Length mm	Offset (mm)			Global (mm)		
					DX	DY	DZ	X	Y	Z
Y05	Y02	Bend	24C70	100.00	0.00	0.00	1200.00	16412.20	7500.00	-19350.00
Y02	Y03	Run	24C70	266.67	3200.00	0.00	0.00	19612.20	7500.00	-19350.00
Y03	Y06	Run	24C70	321.67	3860.00	0.00	0.00	23472.20	7500.00	-19350.00
Y06	Y04	Run	24C70	321.67	3860.00	0.00	0.00	27332.20	7500.00	-19350.00
*** Segment Z										
Origin	Y00	Point	24C70	0.00	16412.20	7500.00	-24350.00	16412.20	7500.00	-24350.00
Y00	Z01	Junction	24C70	108.33	0.00	0.00	-1300.00	16412.20	7500.00	-25650.00
Z01	Z02	Run	24C70	108.33	0.00	0.00	-1300.00	16412.20	7500.00	-26950.00
*** Segment AA										
Origin	J22	Point	4"B06	0.00	20341.26	10133.91	-10500.00	20341.26	10133.91	-10500.00
J22	AA01	Junction	4"B06	5.58	-66.94	0.00	0.00	20274.31	10133.91	-10500.00
AA01	AA02	Reducer	6"B06	10.00	-120.00	0.00	0.00	20154.31	10133.91	-10500.00
AA02	AA03	Run	6"B06	71.67	-860.00	0.00	0.00	19294.31	10133.91	-10500.00
AA03	AA04	Run	6"B06	38.33	-460.00	0.00	0.00	18834.31	10133.91	-10500.00
AA04	AA05	Valve	6"B06	50.84	-610.11	0.00	0.00	18224.20	10133.91	-10500.00
AA05	AA06	Run	6"B06	33.33	-400.00	0.00	0.00	17824.20	10133.91	-10500.00
AA06	AA07	Tee	6"B06	33.33	-400.00	0.00	0.00	17424.20	10133.91	-10500.00
AA07	AA08	Bend	6"B06	33.33	-400.00	0.00	0.00	17024.20	10133.91	-10500.00
AA08	AW01	Tee	6"B06	33.33	0.00	-400.00	0.00	17024.20	9733.91	-10500.00
*** Segment AB										
Origin	J18	Tee	4"C06	0.00	20541.25	9676.95	-10500.00	20541.25	9676.95	-10500.00
J18	AB01	Bend	4"C06	25.00	0.00	0.00	300.00	20541.25	9676.95	-10200.00
AB01	AB02	Run	4"C06	79.17	-950.00	0.00	0.00	19591.26	9676.95	-10200.00
AB02	AB03	Valve	4"C06	38.08	-456.95	0.00	0.00	19134.31	9676.95	-10200.00
AB03	AB04	Valve	4"C06	38.10	-457.20	0.00	0.00	18677.11	9676.95	-10200.00
*** Segment AD										

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X	Y	Z
Origin	AB04	Point	4"B06	0.00	18677.11	9676.95	-10200.00	18677.11	9676.95	-10200.00
AB04	AD01	Junction	4"B06	52.20	-626.46	0.00	0.00	18050.65	9676.95	-10200.00
AD01	AD02	Bend	4"B06	52.20	-626.45	0.00	0.00	17424.20	9676.95	-10200.00
AD02	AD03	Bend	4"B06	38.08	0.00	456.96	0.00	17424.20	10133.91	-10200.00
AD03	AA07	Tee	4"B06	25.00	0.00	0.00	-300.00	17424.20	10133.91	-10500.00
*** Segment AE										
Origin	O03	Point	4"C06	0.00	44191.20	7500.00	-8250.00	44191.20	7500.00	-8250.00
O03	AE01	Run	4"C06	58.33	0.00	700.00	0.00	44191.20	8200.00	-8250.00
*** Segment AF										
Origin	AE01	Point	4"C06	0.00	44191.20	8200.00	-8250.00	44191.20	8200.00	-8250.00
AE01	AF01	Junction	4"C06	1.67	0.00	20.00	0.00	44191.20	8220.00	-8250.00
AF01	AF02	Valve	4"C06	38.08	0.00	456.95	0.00	44191.20	8676.95	-8250.00
AF02	AF03	Bend	4"C06	25.00	0.00	300.00	0.00	44191.20	8976.95	-8250.00
AF03	AF04	Run	4"C06	60.00	0.00	0.00	720.00	44191.20	8976.95	-7530.00
AF04	AF05	Run	4"C06	223.33	0.00	0.00	2680.00	44191.20	8976.95	-4850.00
AF05	AF06	Bend	4"C06	66.67	0.00	0.00	800.00	44191.20	8976.95	-4050.00
AF06	AF07	Run	4"C06	158.33	-1900.00	0.00	0.00	42291.20	8976.95	-4050.00
AF07	AF08	Run	4"C06	333.33	-4000.00	0.00	0.00	38291.20	8976.95	-4050.00
AF08	AF09	Run	4"C06	375.00	-4500.00	0.00	0.00	33791.20	8976.95	-4050.00
AF09	AF27	Run	4"C06	372.63	-4471.50	0.00	0.00	29319.70	8976.95	-4050.00
AF27	AF10	Bend	4"C06	372.62	-4471.50	0.00	0.00	24848.20	8976.95	-4050.00
AF10	AF11	Run	4"C06	250.00	0.00	0.00	-3000.00	24848.20	8976.95	-7050.00
AF11	AF12	Bend	4"C06	333.33	0.00	0.00	-4000.00	24848.20	8976.95	-11050.00
AF12	AF13	Run	4"C06	62.50	-750.00	0.00	0.00	24098.20	8976.95	-11050.00
AF13	AF14	Bend	4"C06	133.33	-1600.00	0.00	0.00	22498.20	8976.95	-11050.00
AF14	AF15	Run	4"C06	50.00	0.00	0.00	-600.00	22498.20	8976.95	-11650.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm)		Z
								X	Y	
AF15	AF16	Bend	4"C06	33.33	0.00	0.00	-400.00	22498.20	8976.95	-12050.00
AF16	AF17	Bend	4"C06	58.33	0.00	700.00	0.00	22498.20	9676.95	-12050.00
AF17	AF18	Run	4"C06	25.00	-300.00	0.00	0.00	22198.20	9676.95	-12050.00
AF18	AF19	Run	4"C06	8.33	-100.00	0.00	0.00	22098.20	9676.95	-12050.00
AF19	AF20	Valve	4"C06	38.08	-456.95	0.00	0.00	21641.26	9676.95	-12050.00
AF20	AF21	Run	4"C06	75.00	-900.00	0.00	0.00	20741.25	9676.95	-12050.00
AF21	AF22	Tee	4"C06	16.67	-200.00	0.00	0.00	20541.25	9676.95	-12050.00
AF22	AF23	Bend	4"C06	16.67	-200.00	0.00	0.00	20341.25	9676.95	-12050.00
AF23	AF24	Run	4"C06	16.67	0.00	200.00	0.00	20341.26	9876.95	-12050.00
AF24	AF25	Valve	4"C06	19.04	0.00	228.48	0.00	20341.26	10105.43	-12050.00
AF25	AF26	Run	4"C06	2.37	0.00	28.48	0.00	20341.26	10133.91	-12050.00
*** Segment AI										
Origin	AF26	Point	4"B06	0.00	20341.26	10133.91	-12050.00	20341.26	10133.91	-12050.00
AF26	AI01	Junction	4"B06	5.58	-66.94	0.00	0.00	20274.31	10133.91	-12050.00
AI01	AI02	Reducer	6"B06	10.00	-120.00	0.00	0.00	20154.31	10133.91	-12050.00
AI02	AI03	Run	6"B06	71.67	-860.00	0.00	0.00	19294.31	10133.91	-12050.00
AI03	AI04	Run	6"B06	38.33	-460.00	0.00	0.00	18834.31	10133.91	-12050.00
AI04	AI05	Valve	6"B06	50.84	-610.11	0.00	0.00	18224.20	10133.91	-12050.00
AI05	AI06	Run	6"B06	33.33	-400.00	0.00	0.00	17824.20	10133.91	-12050.00
AI06	AI07	Tee	6"B06	33.33	-400.00	0.00	0.00	17424.20	10133.91	-12050.00
AI07	AI08	Bend	6"B06	33.33	-400.00	0.00	0.00	17024.20	10133.91	-12050.00
AI08	AW02	Tee	6"B06	33.33	0.00	-400.00	0.00	17024.20	9733.91	-12050.00
*** Segment AJ										
Origin	AF22	Tee	4"C06	0.00	20541.25	9676.95	-12050.00	20541.25	9676.95	-12050.00
AF22	AJ01	Bend	4"C06	25.00	0.00	0.00	300.00	20541.25	9676.95	-11750.00
AJ01	AJ02	Run	4"C06	79.17	-950.00	0.00	0.00	19591.26	9676.95	-11750.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X Y Z		
AJ02	AJ03	Valve	4"C06	38.08	-456.95	0.00	0.00	19134.31	9676.95	-11750.00
AJ03	AJ04	Valve	4"C06	38.10	-457.20	0.00	0.00	18677.11	9676.95	-11750.00
*** Segment AK										
Origin	AJ04	Point	4"B06	0.00	18677.11	9676.95	-11750.00	18677.11	9676.95	-11750.00
AJ04	AK01	Junction	4"B06	52.20	-626.46	0.00	0.00	18050.65	9676.95	-11750.00
AK01	AK02	Bend	4"B06	52.20	-626.45	0.00	0.00	17424.20	9676.95	-11750.00
AK02	AK03	Bend	4"B06	38.08	0.00	456.96	0.00	17424.20	10133.91	-11750.00
AK03	AI07	Tee	4"B06	25.00	0.00	0.00	-300.00	17424.20	10133.91	-12050.00
*** Segment AR										
Origin	T07	Tee	6"C06	0.00	49620.20	7500.00	-19350.00	49620.20	7500.00	-19350.00
T07	AR01	Bend	6"C06	123.08	0.00	1477.00	0.00	49620.20	8977.00	-19350.00
AR01	AR02	Run	6"C06	83.33	0.00	0.00	1000.00	49620.20	8977.00	-18350.00
AR02	AR03	Bend	6"C06	228.08	0.00	0.00	2737.00	49620.20	8977.00	-15613.00
AR03	AR04	Run	6"C06	133.33	-1600.00	0.00	0.00	48020.21	8977.00	-15613.00
AR04	AR05	Run	6"C06	333.33	-4000.00	0.00	0.00	44020.21	8977.00	-15613.00
AR05	AR06	Run	6"C06	500.00	-6000.00	0.00	0.00	38020.21	8977.00	-15613.00
AR06	AR22	Run	6"C06	376.10	-4513.25	0.00	0.00	33506.96	8977.00	-15613.00
AR22	AR21	Run	6"C06	376.10	-4513.25	0.00	0.00	28993.71	8977.00	-15613.00
AR21	AR07	Run	6"C06	335.54	-4026.50	0.00	0.00	24967.21	8977.00	-15613.00
AR07	AR08	Bend	6"C06	72.42	-869.00	0.00	0.00	24098.21	8977.00	-15613.00
AR08	AR09	Bend	6"C06	163.58	0.00	0.00	1963.00	24098.21	8977.00	-13650.00
AR09	AR10	Bend	6"C06	83.33	-1000.00	0.00	0.00	23098.21	8977.00	-13650.00
AR10	AR11	Bend	6"C06	58.33	0.00	700.00	0.00	23098.21	9677.00	-13650.00
AR11	AR12	Tee	6"C06	33.33	-400.00	0.00	0.00	22698.21	9677.00	-13650.00
AR12	AR13	Run	6"C06	25.00	-300.00	0.00	0.00	22398.21	9677.00	-13650.00
AR13	AR14	Valve	6"C06	50.84	-610.11	0.00	0.00	21788.10	9677.00	-13650.00

From	To	Comp.	Pipe ID	Length mm	Offset (mm)			Global (mm)		
					DX	DY	DZ	X	Y	Z
AR14	AR15	Run	6"C06	45.00	-540.00	0.00	0.00	21248.10	9677.00	-13650.00
AR15	AR16	Run	6"C06	45.00	-540.00	0.00	0.00	20708.10	9677.00	-13650.00
AR16	AR17	Valve	6"C06	50.84	-610.11	0.00	0.00	20097.99	9677.00	-13650.00
AR17	AR18	Bend	6"C06	37.50	-450.00	0.00	0.00	19647.99	9677.00	-13650.00
AR18	AR19	Run	6"C06	25.00	0.00	300.00	0.00	19647.99	9977.00	-13650.00
AR19	AR20	Valve	6"C06	19.08	0.00	229.00	0.00	19647.99	10206.00	-13650.00
*** Segment AS										
Origin	AR20	Point	6"B06	0.00	19647.99	10206.00	-13650.00	19647.99	10206.00	-13650.00
AR20	AS01	Junction	6"B06	29.17	-350.00	0.00	0.00	19297.99	10206.00	-13650.00
AS01	AS02	Run	6"B06	29.17	-350.00	0.00	0.00	18947.99	10206.00	-13650.00
AS02	AS03	Valve	6"B06	50.84	-610.11	0.00	0.00	18337.88	10206.00	-13650.00
AS03	AS04	Bend	6"B06	109.47	-1313.68	0.00	0.00	17024.20	10206.00	-13650.00
AS04	AW04	Tee	6"B06	39.34	0.00	-472.09	0.00	17024.20	9733.91	-13650.00
*** Segment AT										
Origin	AR12	Tee	6"C06	0.00	22698.21	9677.00	-13650.00	22698.21	9677.00	-13650.00
AR12	AT01	Bend	6"C06	55.83	0.00	0.00	-670.00	22698.21	9677.00	-14320.00
AT01	AT02	Run	6"C06	20.83	-250.00	0.00	0.00	22448.21	9677.00	-14320.00
AT02	AT03	Run	6"C06	25.00	-300.00	0.00	0.00	22148.21	9677.00	-14320.00
AT03	AT04	Valve	6"C06	50.84	-610.11	0.00	0.00	21538.10	9677.00	-14320.00
AT04	AT05	Run	6"C06	24.17	-290.00	0.00	0.00	21248.10	9677.00	-14320.00
AT05	AT06	Run	6"C06	45.00	-540.00	0.00	0.00	20708.10	9677.00	-14320.00
AT06	AT07	Valve	6"C06	50.84	-610.11	0.00	0.00	20097.99	9677.00	-14320.00
AT07	AT08	Bend	6"C06	37.50	-450.00	0.00	0.00	19647.99	9677.00	-14320.00
AT08	AT09	Run	6"C06	25.00	0.00	300.00	0.00	19647.99	9977.00	-14320.00
AT09	AT10	Valve	6"C06	19.08	0.00	229.00	0.00	19647.99	10206.00	-14320.00
*** Segment AV										

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X	Y	Z
Origin	AT10	Point	6"B06	0.00	19647.99	10206.00	-14320.00	19647.99	10206.00	-14320.00
AT10	AV01	Junction	6"B06	29.17	-350.00	0.00	0.00	19297.99	10206.00	-14320.00
AV01	AV02	Run	6"B06	29.17	-350.00	0.00	0.00	18947.99	10206.00	-14320.00
AV02	AV03	Valve	6"B06	50.84	-610.11	0.00	0.00	18337.88	10206.00	-14320.00
AV03	AV04	Bend	6"B06	109.47	-1313.68	0.00	0.00	17024.20	10206.00	-14320.00
AV04	AW05	Tee	6"B06	39.34	0.00	-472.09	0.00	17024.20	9733.91	-14320.00
*** Segment AW										
Origin	AW00	Point	6"B06	0.00	17024.20	9733.91	-10200.00	17024.20	9733.91	-10200.00
AW00	AW01	Tee	6"B06	25.00	0.00	0.00	-300.00	17024.20	9733.91	-10500.00
AW01	AW02	Tee	6"B06	129.17	0.00	0.00	-1550.00	17024.20	9733.91	-12050.00
AW02	AW03	Run	6"B06	44.83	0.00	0.00	-538.00	17024.20	9733.91	-12588.00
AW03	AW04	Tee	6"B06	88.50	0.00	0.00	-1062.00	17024.20	9733.91	-13650.00
AW04	AW05	Tee	6"B06	55.83	0.00	0.00	-670.00	17024.20	9733.91	-14320.00
AW05	AW06	Bend	6"B06	91.67	0.00	0.00	-1100.00	17024.20	9733.91	-15420.00
AW06	AW07	Bend	6"B06	96.67	0.00	-1160.00	0.00	17024.20	8573.91	-15420.00
AW07	AW08	Run	6"B06	66.67	-800.00	0.00	0.00	16224.20	8573.91	-15420.00
AW08	AW09	Run	6"B06	358.33	-4300.00	0.00	0.00	11924.20	8573.91	-15420.00
AW09	AW10	Bend	6"B06	100.00	-1200.00	0.00	0.00	10724.20	8573.91	-15420.00
AW10	AW11	Run	6"B06	166.67	0.00	0.00	2000.00	10724.20	8573.91	-13420.00
AW11	AW12	Bend	6"B06	166.67	0.00	0.00	2000.00	10724.20	8573.91	-11420.00
AW12	AW13	Run	6"B06	333.33	0.00	-4000.00	0.00	10724.20	4573.91	-11420.00
*** Segment AX										
Origin	AX00	Point	6"B06	0.00	10724.20	4573.91	-8820.00	10724.20	4573.91	-8820.00
AX00	AX01	Run	6"B06	1.67	0.00	20.00	0.00	10724.20	4593.91	-8820.00
AX01	AX02	Valve	6"B06	38.10	0.00	457.20	0.00	10724.20	5051.11	-8820.00
AX02	AX03	Bend	6"B06	293.57	0.00	3522.80	0.00	10724.20	8573.91	-8820.00

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X	Y	Z
AX03	AX04	Run	6"B06	100.00	0.00	0.00	1200.00	10724.20	8573.91	-7620.00
AX04	AX05	Run	6"B06	450.00	0.00	0.00	5400.00	10724.20	8573.91	-2220.00
AX05	AX06	Run	6"B06	500.00	0.00	0.00	6000.00	10724.20	8573.91	3780.00
AX06	AX07	Run	6"B06	141.67	0.00	0.00	1700.00	10724.20	8573.91	5480.00
*** Segment AY										
Origin	O02	Point	8"C03	0.00	48391.20	7500.00	-8250.00	48391.20	7500.00	-8250.00
O02	AY01	Run	8"C03	58.33	0.00	700.00	0.00	48391.20	8200.00	-8250.00
AY01	AY02	Valve	8"C03	61.43	0.00	737.11	0.00	48391.20	8937.11	-8250.00
AY02	AY03	Bend	8"C03	66.76	0.00	801.09	0.00	48391.20	9738.20	-8250.00
AY03	AY04	Bend	8"C03	141.67	0.00	0.00	-1700.00	48391.20	9738.20	-9950.00
AY04	AY05	Bend	8"C03	186.52	0.00	-2238.20	0.00	48391.20	7500.00	-9950.00
AY05	AY06	Run	8"C03	70.83	-850.00	0.00	0.00	47541.20	7500.00	-9950.00
AY06	AY07	Run	8"C03	83.33	-1000.00	0.00	0.00	46541.20	7500.00	-9950.00
AY07	AY08	Valve	8"C03	61.43	-737.11	0.00	0.00	45804.09	7500.00	-9950.00
AY08	AY09	Bend	8"C03	159.99	-1919.89	0.00	0.00	43884.20	7500.00	-9950.00
AY09	Q03	Tee	8"C03	108.33	0.00	0.00	-1300.00	43884.20	7500.00	-11250.00

